

LEROY AI // SYSTEM DOCUMENTATION & SETUP GUIDE

Revenue Optimization & Employee Knowledge Platform (FastAPI + React + LangGraph + Supabase pgvector)

1. Executive Overview

Leroy AI is an enterprise executive copilot platform designed for real-time revenue strategy optimization and human capital intelligence. Built with a high-density **Bloomberg Terminal UI**, it orchestrates multi-agent RAG pipelines using **LangGraph**, **LangChain**, **Groq LLaMA-3.3 LLM Engine**, and **Supabase pgvector**.

2. Technical Stack Specifications

Component	Technology / Library	Role / Purpose
Frontend UI	React 18, Vite, Lucide Icons, Recharts	Bloomberg Terminal theme with dark slate grid & live marquee
Backend Framework	FastAPI (Python 3.13), Uvicorn	REST API endpoints for chat, revenue, employees, knowledge
Agentic RAG	LangChain, LangGraph, Groq LLaMA-3.3-70B	Router, vector retriever, telemetry calculator & response generator
Vector Database	Supabase PostgreSQL + pgvector extension	384-dimensional vector similarity store with match_documents RPC
Embeddings Engine	SentenceTransformers (all-MiniLM-L6-v2)	384-d dense vector embeddings generation for RAG documents

3. Step-by-Step Supabase Setup Guide

Step 3.1: Create a Free Supabase Project

- Go to <https://supabase.com> and sign in.
- Click **New Project**, select your organization, set a database password, and create the project.

Step 3.2: Obtain Supabase Credentials

- In your project dashboard, navigate to **Project Settings > API**.
- Copy your **Project URL** (e.g., `https://xyz.supabase.co`).
- Copy your **anon public API Key** (starts with `eyJ...`).

Step 3.3: Execute pgvector SQL Migration

- In Supabase Dashboard, click **SQL Editor** on the left menu.
- Open the project file `backend/supabase_schema.sql` and paste its contents into the SQL Editor.
- Click **Run**. This will enable the pgvector extension, create tables (`employees`, `revenue_metrics`, `knowledge_documents`), and register the vector search function `match_documents`.

```
-- Supabase SQL Schema Snippet --
CREATE EXTENSION IF NOT EXISTS vector;
CREATE TABLE IF NOT EXISTS knowledge_documents (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title VARCHAR(255) NOT NULL,
  category VARCHAR(100) NOT NULL,
  content TEXT NOT NULL,
  embedding vector(384)
);
```

4. Step-by-Step Groq API Setup Guide

Step 4.1: Get a Free Groq API Key

1. Go to <https://console.groq.com> and log in.
2. Click **API Keys** in the sidebar menu.
3. Click **Create API Key**, give it a name (e.g. leroy-ai), and copy the generated key (starts with gsk ...).

5. Environment Configuration (.env)

Open `e:\Production RAG\backend\.env` and populate your credentials:

```
# Leroy AI Production Credentials  
GROQ_API_KEY=gsk_your_actual_groq_api_key_here  
GROQ_MODEL=llama-3.3-70b-versatile  
  
SUPABASE_URL=https://your-project.supabase.co  
SUPABASE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.your_supabase_anon_key  
  
PORT=8000  
HOST=0.0.0.0
```

6. Launching Leroy AI

Run Backend Server:

```
cd "e:\Production RAG\backend"
python -m uvicorn main:app --port 8000 --reload
```

Run Frontend UI:

```
cd "e:\Production RAG\frontend"
npx vite --port 3000
```